MSME DIGITAL LENDING

First-time tester guide

Borrower registration to human-approved mock sanction

Version 0.2 - revised synthetic journey

Audience: a tester who has never used this application. Follow the walkthrough first, then execute the 40 checks and record results.

All synthetic. OTP, KYC, bank statements, GST, ITR, bureau records, Udyam and CBS posting are simulated. No real identity details, bank credentials or money are required.

Approval always requires a person. Analysis, a passing bureau score or an evidence graph is not an approval. Only the branch manager can issue the final decision.

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This edition replaces the upload-first guide for new borrower applications. Old officer-created cases still use the legacy workflow.

1. Prepare the environment

Ask the project owner for the UI URL and the two staff tokens. Never paste tokens into a defect report. Default UI: http://localhost:8080.

Existing installation: back up the project and database, stop containers, replace source files, and retain the existing .env and Docker volumes. Do not replace the database password.

Fresh installation: copy .env.example to .env once. No external provider keys are needed. Keep LLM_MODE=mock.

```
docker compose up --build -d
docker compose ps
```

Expect db and api Healthy, client running. The first 100 provider profiles are created automatically. No seed command is needed for borrower applications.

Role	How to sign in	Main responsibility
Borrower	Select synthetic profile; use displayed OTP	Apply, consent, fetch, supply evidence and track
Credit officer	Credit team portal; ANALYST_TOKEN	Review, verify, request evidence and structure terms
Branch manager	Credit team portal; REVIEWER_TOKEN	Approve rate exceptions and final sanction

To increase profile count (idempotently):

```
docker compose exec api python generate_profiles.py --count 500
```

Use 100-500. Existing profiles and applications remain. Profile scenario pattern repeats every six IDs. Prefer a fresh application for each destructive decision test.

Suggested test records: SYN-0001 consistent; 0002 low bureau; 0003 missing GST; 0004 low capacity; 0005 GST mismatch; 0006 balance drop.

2. Borrower: first application

1. Register / sign in

Select SYN-0001, Synthetic Owner 001. Click Get demo OTP. Enter the code shown and verify. No SMS is sent.

2. Verify KYC

Keep the displayed PAN DEMOPAN0001 and date of birth 1988-01-15. Request a separate KYC OTP, enter it and continue. Name comes from the matched profile.

3. Review business and consent

Review the synthetic Udyam, GST and sector. Select all five consents: KYC, bank, tax, bureau and processing.

4. Configure loan

Select Term loan. Enter INR 1,000,000, 36 months and a synthetic equipment purchase purpose.

5. Submit and fetch

Click Submit & fetch synthetic data. Wait for processing. The application and source cards appear; bank, GST, ITR, financials, KYC, bureau and Udyam should be available.

6. Track next actions

Expect awaiting_review, with a HUMAN_VERIFICATION requirement. This is normal. Read requests and outstanding checks. The borrower cannot approve the case.

Record: profile ID, application ID and current revision. Use these to find the same case in the credit team portal.

3. Credit review and sanction

1. Open the case as officer

Choose Credit team portal, enter ANALYST_TOKEN, search DEMOPAN0001 and open the correct application. The case tabs are separate from sidebar Overview.

2. Complete Verification & terms

Tick identity, business nature and use-of-funds verification. Record simulated interview notes and market/viability assessment. Save, then click Run analysis.

3. Check financials and evidence

For the default case at 14%: monthly operating cash INR 220,000; existing monthly debt INR 20,000; EMI INR 34,177.63 (UI may round); DSCR about 4.06. Follow citations and review all policy outcomes.

4. Review terms and CAM

Changing amount, tenure or rate invalidates results. Save and rerun. Any change from base rate needs manager approval. A rate exception approval also requires CAM regeneration.

5. Switch to branch manager

Sign out of staff view and sign in using REVIEWER_TOKEN. Resolve pending rate decisions, rerun if necessary, inspect the current CAM, then Human review > approve with comments.

6. Confirm the result

Mock ledger has exactly one row for the application. Return to Borrower portal, Track application, Refresh status. Expect approved and mock ledger posted. No funds move.

4. Test cases 01-04

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J01 - Wrong login OTP

Action: Wrong login OTP.

Expected: Error; no borrower session.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J02 - Reuse consumed OTP

Action: Reuse consumed OTP.

Expected: Rejected.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J03 - Wait over 5 minutes before OTP verification

Action: Wait over 5 minutes before OTP verification.

Expected: Expired OTP rejected; resend needed.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J04 - Five wrong attempts, then correct OTP

Action: Five wrong attempts, then correct OTP.

Expected: Challenge remains blocked.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 05-08

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J05 - Request more than five OTPs for one mobile in one minute

Action: Request more than five OTPs for one mobile in one minute.

Expected: Rate limited; retry after one minute.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J06 - Change PAN/DOB from selected profile before submission

Action: Change PAN/DOB from selected profile before submission.

Expected: Submission rejected, no identity mismatch case created.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J07 - Leave one consent unchecked

Action: Leave one consent unchecked.

Expected: Cannot continue; direct API submission also rejected.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J08 - Complete the successful flow above

Action: Complete the successful flow above.

Expected: One approved case and one mock ledger row.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 09-12

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J09 - Officer attempts final approve/reject

Action: Officer attempts final approve/reject.

Expected: Disabled UI and server 403.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J10 - Manager attempts approval before interview

Action: Manager attempts approval before interview.

Expected: Blocked by HUMAN_VERIFICATION.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J11 - Change amount or tenure

Action: Change amount or tenure.

Expected: Revision increases; old CAM cannot approve; financials must rerun.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J12 - Change rate 14 to 13.5

Action: Change rate 14 to 13.5.

Expected: Pending manager rate exception; recalculated EMI lower; approval blocked.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 13-16

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J13 - Officer attempts rate exception approval

Action: Officer attempts rate exception approval.

Expected: Server 403.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J14 - Manager approves rate exception

Action: Manager approves rate exception.

Expected: Rate approved, CAM invalidated; rerun then inspect before final approval.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J15 - Manager rejects exception

Action: Manager rejects exception.

Expected: RATE_EXCEPTION remains blocking; officer must revise terms.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J16 - Rate below 13 or above 19 with default base 14

Action: Rate below 13 or above 19 with default base 14.

Expected: Rejected by demo hard bounds.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 17-20

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J17 - Change terms after approved exception

Action: Change terms after approved exception.

Expected: New rate decision required for a non-base rate.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J18 - Choose LAP/construction; omit survey/title/margin

Action: Choose LAP/construction; omit survey/title/margin.

Expected: PROPERTY gate blocks final approval.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J19 - LAP with value 2,000,000, margin 25%, loan 1,000,000; add surveyor/report/title

Action: LAP with value 2,000,000, margin 25%, loan 1,000,000; add surveyor/report/title.

Expected: PROPERTY passes after verification and analysis.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J20 - LAP amount exceeds value minus margin

Action: LAP amount exceeds value minus margin.

Expected: PROPERTY remains blocked.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 21-24

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J21 - SYN-0002

Action: SYN-0002.

Expected: Bureau 650, BUREAU check blocks approval; no automatic rejection.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J22 - SYN-0003

Action: SYN-0003.

Expected: GST requires_action; needs_evidence with specific missing GST request.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J23 - Upload legacy 01_consistent/gst.json as missing GST for SYN-0003

Action: Upload legacy 01_consistent/gst.json as missing GST for SYN-0003.

Expected: GST gap resolves after automatic rerun; human verification still required.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J24 - SYN-0004

Action: SYN-0004.

Expected: Insufficient repayment capacity; DSCR fails.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 25-28

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J25 - SYN-0005

Action: SYN-0005.

Expected: GST reconciliation conflict; final approval blocked.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J26 - SYN-0006

Action: SYN-0006.

Expected: Balance-drop observation cites source transactions; investigation required.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J27 - Officer requests exact missing evidence

Action: Officer requests exact missing evidence.

Expected: Borrower sees request; current CAM invalidated.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J28 - Borrower submits clarification text

Action: Borrower submits clarification text.

Expected: Audit records response; request does not silently resolve.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 29-32

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J29 - Staff reviews evidence, resolves request and reruns

Action: Staff reviews evidence, resolves request and reruns.

Expected: Request resolved; new CAM includes resolution.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J30 - Retry provider fetch on editable case

Action: Retry provider fetch on editable case.

Expected: Existing active evidence not duplicated or overwritten.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J31 - Upload identical file again

Action: Upload identical file again.

Expected: Duplicate rejected.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J32 - Replace document explicitly

Action: Replace document explicitly.

Expected: Old version retained; prior CAM stale; new analysis uses replacement.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 33-36

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J33 - Log in as another borrower

Action: Log in as another borrower.

Expected: Only own cases appear; cross-case API requests return 404.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J34 - Repeat final approval

Action: Repeat final approval.

Expected: Rejected; ledger remains one row.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J35 - Edit/fetch/upload after approval or rejection

Action: Edit/fetch/upload after approval or rejection.

Expected: Blocked; final case frozen.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J36 - Reload or restart containers

Action: Reload or restart containers.

Expected: Profiles/cases/approvals retained; valid session can be reused.

Result: _____ Case / revision: _____

Actual result / evidence: _____

4. Test cases 37-40

Run against a fresh or suitable editable case unless this check explicitly requires a final case. Default rate is 14%. Record Pass, Fail, Blocked or Not run.

J37 - Borrower sign out; reuse old session token

Action: Borrower sign out; reuse old session token.

Expected: Session rejected.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J38 - Invalid staff token

Action: Invalid staff token.

Expected: Staff login denied.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J39 - Narrow mobile viewport

Action: Narrow mobile viewport.

Expected: Forms and buttons usable; no page-wide horizontal overflow.

Result: _____ Case / revision: _____

Actual result / evidence: _____

J40 - Print current CAM

Action: Print current CAM.

Expected: Financials, source references, human inputs and current terms readable.

Result: _____ Case / revision: _____

Actual result / evidence: _____

5. Troubleshooting

“I uploaded everything, but it still needs evidence.”

Open the case-level Overview and read the exact Evidence requests. Inspect Documents for failed parsing or missing pages, then Analysis for policy failures. A file being present does not mean its contents are complete, readable, matched to this borrower or from the correct reporting period.

Symptom	What to check / do
KYC mismatch	New profile names differ from the old Asha fixture. Keep provider KYC or replace with matching synthetic data. Do not add “Applicant:” to a name.
Missing GST (SYN-0003)	Upload fixtures/synthetic_cases/01_consistent/gst.json as GST. The API reruns analysis. Staff verification is still required.
Open officer request	Borrower clarification text does not close it. Staff must review, resolve in Human review and rerun.
Current CAM missing	Run analysis after terms, verification, rate approval or evidence changes. Old CAMs cannot approve.
LAP / construction blocked	Record surveyor, report reference, clear title, property value and 10-40% margin. Loan must fit value minus margin.
Approval disabled	Check role, current revision/CAM, every policy check and unresolved requests. Awaiting review is not approvable.

What this version simulates

Calls are manually recorded simulated interviews, not a live video agent. Provider fetching uses local adapters, not a deployed bank MCP server. MSME category is fixture metadata, not statutory certification. The bounded orchestrator uses sequential checkpoints; document replacement reruns extraction and downstream nodes. No learned cross-case memory or real disbursement is included.

6. Results and handover

Tester: _____ Date: _____
Build/version: _____ Browser: _____

For every check record the test ID, application ID/revision, actual behavior and evidence. Use a separate sheet if needed. Do not mark blocked or unexecuted tests as passed.

Test IDs	Pass / Fail / Blocked / Not run	Evidence reference
J01-J04		
J05-J08		
J09-J12		
J13-J16		
J17-J20		
J21-J24		
J25-J28		
J29-J32		
J33-J36		
J37-J40		

Defect report template

Title / severity: _____
Role (no token): _____ Profile: _____ Test ID: _____
Application ID / revision: _____
Reproduction steps: _____
Expected / actual: _____
Screenshot or API error + timestamp: _____

Release handover

Treat cross-borrower access, officer final approval, sanction with unresolved gates, stale-CAM approval or duplicate ledger entries as critical failures. Record open defects and retest fixes. Confirm the happy path, missing evidence, negotiated rate and property flows before sign-off.

Automated checks executed during development: 32 backend tests, two frontend tests, production frontend build, and a Chromium borrower-to-manager flow including mobile layout. This is not a substitute for recording your own manual test results.